

MOU PAUL

Engineering Leader experienced in the LLM pre-training data lifecycle. Architected Petabyte-scale Spark/Ray pipelines to process trillions of tokens, automated high-fidelity curation via LLM-as-a-Judge

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Technical Skills

Language Python Golang Typescript/Javascript Rust
Platform Kubernetes Docker Spark BigQuery BigTable Node.js gRPC
Infra Kafka Consul Vault Nginx Redis Sqlite
Library Ray Pytorch Rocksdb Electron Next.js React

Selected Projects

Learning about CT with Gemini
<https://go.mou.dev/ct-voxel>

Listening with LLM
<https://go.mou.dev/blog-llm>

Professional Experiences

Member of Technical Staff @ Microsoft AI, Redmond 2024/12-Present

- Architected petabyte-scale Spark/Ray data pipelines to process trillions of tokens from 3B+ unstructured PDFs for LLM pre-training, utilizing ablation studies to refine data-mixture strategies
- Engineered automated filtering and deduplication workflows using vector embeddings and LLM-as-a-Judge to curate high-fidelity training sets from raw, unstructured data

Software Engineer @ Roblox, San Mateo 2022/05-2024/12

- Key technical contributor and advocate for driving datalake schematization at Roblox, delivering over \$10M savings in annual cloud costs
- Promoted DevOps best practices via code-based dashboard and alert generation, runbook documentation, capacity planning, and incident postmortems

Software Engineer @ Twitter, San Francisco 2018/06-2022/04

- Scaled anti-spam infrastructure 10x post-acquisition by re-architecting the Smyte anti-spam engine to support Twitter-scale traffic
- Led dataset migration to BigQuery within Twitter Health to accelerate cloud adoption across the org
- Hardened data privacy for GDPR compliance by designing a personal data anonymizer system for the anti-spam infrastructure

Software Engineer @ Smyte, San Francisco 2015/11-2018/06

- Core member of four engineers building and scaling Smyte's anti-spam engine protecting hundreds of millions of users
- Developed and maintained automation and monitoring for multiple kubernetes cluster with thousands of pods
- Contributed to high impact initiatives such as GDPR compliance and building first machine learning stack for abusive image detection

Educations

Udacity
Self-Driving Car Engineer I
Certificate

Stanford University
Computer Science
Master (withdrew)

University of Waterloo
System Design
Bachelor